INCH-POUND
A-A-59187
9 September 1998

COMMERCIAL ITEM DESCRIPTION

RANGE, CONVECTION OVEN, ELECTRIC, MODULAR (NAVAL SHIPBOARD)

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE

This commercial item description (CID) covers modular bolted-construction electric ranges that have convection ovens with griddle tops and hot plates. The ranges are intended for Naval shipboard use in galleys for high speed griddle cooking and forced air convection oven cooking.

2. CLASSIFICATION

Style 1: Range with griddle top
Style 2: Range with griddle top and French hot plates
Style 3: Range with six hot plates (French or regular)

Size 1: 36 inches wide by 38 inches deep by 30 inches high (surface ship)
Size 2: 42 inches wide by 28 inches deep by 46 inches high (submarine)

3. SALIENT CHARACTERISTICS

3.1 Design and construction. The range shall be a free standing, heavy duty unit, provided with electrically heated griddle top, hot plates, and convection oven. The oven shall have blowers for forced air circulation throughout the compartment. The oven shall have tubular, clamp-on, Calrod-type heating elements located behind removable protective baffle plates on the sides, back wall, or top of the oven compartment interior, to produce an even temperature distribution throughout the oven. The oven shall be provided with shelf supports. The range shall be provided with a vent, actuated by a manual control on the front or top of the range. Electrical controls shall be located on a setback. The range shall be stainless steel except for the griddle top, motor, and motor housing, miscellaneous hardware, and wiring. The range frame shall be angle or reinforced sheet metal construction with

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joints firmly secured and shall adequately support a uniform load of 75 pounds per square foot when tested. The thickness of the range front, sides, and back shall be not less than 20-gauge stainless steel. The range shall be provided in modules or finished sections which can be assembled and bolted together at the installation site to form a complete and usable convection oven range. Those modules or sections requiring electricity shall be completely wired and equipped with the necessary interconnections such as terminal blocks, quick-connect wiring harnesses, or connector plugs, appropriately marked for proper connection. When connected together they shall create the electrical wiring of a fully assembled unit. The range, when assembled, shall be of rigid and durable construction. (Shipboard requirement, see 7.2)

3.1.1 Style 1, size 1 range. The style 1, size 1 range griddle top shall be 36 inches wide by 24 inches deep.

3.1.2 Style 1, size 2 range. The style 1, size 2 range griddle top shall be 42 inches wide by 23 inches deep, and shall have four thermal areas, each with a thermostatic control.

3.1.3 Style 2, size 1 range. The style 2, size 1 range griddle top shall be 24 x 24 inches and shall be operated by two thermostats. The two hot plates shall each be operated by either an infinite heat switch or a three heat setting/four-position indicating switch (HIGH, MEDIUM, LOW, and OFF). One thermostat shall operate each 12 x 24-inch section of griddle top. The griddle plate shall be provided with side and rear splash guards. The hot plate switches shall be labeled.

3.1.4 Style 2, size 2 range. The style 2, size 2 range griddle top shall be 23 inches wide by 18 inches deep and shall be operated by two thermostats. One thermostat shall operate an area of 9 x 23 inches of the griddle top; the other thermostat shall operate the smaller (2/3 size) griddle area. The griddle plate shall be provided with side and rear splash guards. The two French hot plates shall each be not less than 9 inches in diameter and be operated by either an infinite heat switch or a six heat setting/seven-position indicating switch (HIGH, MEDIUM-HIGH, MEDIUM, MEDIUM-LOW, LOW, WARM, and OFF). The hot plate switches shall be labeled.

3.1.5 Style 3 range. The style 3 range shall have six French hot plates not less than 9 inches in diameter or six regular hot plates not less than 8-1/2 x 9-1/2 inches and shall each be operated by either an infinite heat switch or a three heat setting/four-position indicating switch (HIGH, MEDIUM, LOW, and OFF). The hot plate switches shall be labeled.

3.2 Range dimensions (when assembled). The size 1 range exterior dimensions shall be 36 inches wide by 38 inches deep (excluding the grab rail) by 30 inches high (excluding 6-inch legs and a rear splash guard). The size 2 range exterior dimensions shall be 42 inches wide by 28-1/2 inches deep (excluding the grab rail) by 46 inches high (including a 5-inch base and a rear splash guard). The size 1 range oven cavity shall be not less than 20 inches wide by 27 inches deep by 14 inches high. The size 2 range oven cavity shall have 2 oven compartments, each being 28 inches wide by not less than 19 inches deep by 11 inches high.

3.3 Stainless steel materials. Stainless steel shall be any type 300 series (except 303 exclusively for welding), as specified in ASTM A 167.

3.4 Main power switch and indicator lights. A main power switch shall be provided to energize the heating elements and two indicator lights labeled MAIN POWER ON and OVEN READY.

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3.5 Thermostats. Oven thermostats shall be calibrated at 400°F and the griddle plate thermostats shall be calibrated at 350°F. Thermostat controls shall have an indicator light and shall be marked in increments of 25°F. They shall have a range not less than 150 to 450°F and an OFF position for the range top; and a range not less than 150 to 450°F and an OFF position for the oven. The oven temperature sensing bulb shall have a protective guard and covering on the sensing leads.

3.6 Range tops. The thickness of the portion of the range top in front of the surface heating units shall be not less than 16-gauge stainless steel. The entire range top shall be reinforced and secured to the range body in such a manner as to insure against warping or buckling when heated. The style 1 top shall have two grease chutes; the style 2 top shall have one grease chute. Two removable spillage drawers, with securing devices to prevent them from opening unintentionally, shall be provided with each range. A rear splash guard shall be furnished with each range. The thickness of the rear splash guard shall be not less than 18-gauge stainless steel.

3.7 Top heating surfaces. The griddle surface shall be one-piece stainless steel. Hot tops shall be reinforced underneath to minimize warping, with a not less than 3/8-inch thick ribbing. When test operated, heating surfaces shall not be porous or pitted, and shall be flat to within 1/8 inch.

3.8 Heating elements. Heating elements shall be the tubular type and shall be securely fastened. The elements shall not sag and must be removable through the front of the oven from within the oven cavity. The ends of the heating elements shall be sealed and protected to preclude the entrance of moisture. Leads requiring disconnection from terminal boards or strips outside the oven cavity shall not be used. The insulation resistance of each range at normal operating temperature shall be not less than 1 MΩ and at room temperature shall be not less than 2 MΩ.

3.9 Electrical wiring. Electrical wiring shall be enclosed, adequately secured, and protected against spillage, abrasion by removal of spillage drawers, and mechanical damage. Metal edges, with which the wire may come in contact, shall be rounded or covered. Wiring terminal connections shall be tight and secure. Terminal leads shall be labeled or suitably marked as to their function. A ground strap shall be provided from the range to its mounted support.

3.10 Grab rail. The range shall be provided with a stainless steel grab rail, full width across the front. The grab rail shall extend between 3-1/2 and 4 inches beyond the front of the range, level with the top. The grab rail shall be supported by a sufficient number of brackets and hardware to insure rigidity and facilitate removal.

3.11 Sea rails, style 2 and style 3 ranges. The style 2 and style 3 ranges shall be provided with sea rails, located around the hot plates, to prevent the cookware from sliding off. (Shipboard requirement, see 7.2)

3.12 Oven door. The oven door may be either the drop type or the side-opening type. For size 2 ranges, the drop-type oven door shall be spring counterbalanced, and when fully opened shall lay horizontal and remain level and even with the oven bottom. The door opening shall be the full width and height of the oven interior. The door, when opened, shall be capable of supporting a dead weight load of 50 pounds per square foot without breaking or distortion. The side-opening type door, when fully open, shall remain in the fully open position and shall open flush with the oven interior side walls. The fully open door shall not obstruct the removal of oven shelves and shelf

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supports. Means shall be provided for lubricating the door hinges unless installation of the permanently lubricated type is specified. A positive-shut door latch shall be provided for the oven doors. An electric interlock shall be provided to de-energize the blower and heating elements when the oven door is opened.

3.13 Oven chamber lining. The oven chamber lining shall be stainless steel and shall be welded construction without openings, except as required for blower penetration and oven vent.

3.14 Removable oven lining panel. The removable oven lining panel shall be not less than 20-gauge stainless steel, and shall be used in conjunction with the oven chamber lining. The panel shall be removable without tools if possible, but the use of basic tools (e.g., screwdriver, pliers, and adjustable wrench) is acceptable.

3.15 Mounting. The oven shall be provided with four 6-inch high stainless steel legs suitable for secure attachment to the deck.

3.16 Oven shelves. The oven compartment shall be provided with six removable shelves with positive stops. Spacing between the six shelves shall be not less than 2 inches. The shelf shall be a steel wire rod, steel frame with bar grating, or expanded stainless steel. Each shelf shall be capable of supporting a load of 20 pounds per square foot with no greater than 1/8-inch deflection or permanent deformation. Each loaded shelf shall not bind when placed into, or removed from, the shelf supports. The oven shall be capable of holding six 18 x 26-inch bun pans or three 12 x 20 x 4-inch food service pans.

3.17 Oven blower. The oven blower shall provide even circulation and heat distribution throughout the oven. The blower shall be powered by a 1725 RPM ball bearing motor with not less than 1/3 horsepower. The blower shall de-energize when the oven door is opened.

3.18 Oven heat loss. The energy required to maintain the oven at a set temperature shall be not greater than 59 watts per square foot of the interior oven area.

3.19 Oven preheat time. The oven shall reach 330°F from ambient temperature (approximately 70°F) in not more than 20 minutes.

3.20 Inclined operation and structural integrity. The range shall operate satisfactorily, in accordance with the requirements of this CID (such as the oven door remaining closed to prevent loss of contents), when test-operated for 30 seconds inclined at an angle of 15° (30° when specified for submarines) each side of the vertical, in each of two vertical planes at right angles to each other. The oven shall not deviate or rack from perpendicularly to the deck by more than 1/8 inch when the deck is inclined at an angle of 15° (30° when specified for submarines). (Shipboard requirement, see 7.2)

3.21 Environmental suitability. The unit shall be capable of withstanding ship's vibration and motion. Controls, switches, moving parts, and electrical circuits shall operate under shipboard conditions without malfunction, binding, excessive looseness, or damage, when tested in accordance with MIL-STD-167-1, type I equipment. The unit shall be secured to the test machine in the same manner that it will be secured on shipboard. (Shipboard requirement, see 7.2)

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3.22 Electrical requirements. The unit shall operate on 440 VAC, 60 Hz, three-phase power, as defined in MIL-STD-1399, Section 300, and shall have provisions for making direct (hardwired) connections for electric power (2 conductors) and for equipment grounding (1 conductor). All outermost metallic surfaces shall be grounded via the equipment grounding connection. The grounding resistance between any exposed metallic surfaces and the common ground point shall be not greater than 0.1 ohm. Electrical components, other than the hermetically sealed motor, shall be provided in accordance with NEMA 250, Type 13 or equivalent enclosure protection. Metal parts of electrical components and enclosures shall be inherently corrosion resistant or shall be treated and processed for corrosion resistance in accordance with IEEE Standard 45. (Shipboard requirement, see 7.2)

3.23 Control panel. Controls, switches, and indicator lights shall be located in a control panel at the front of the oven.

3.24 Oven cavity light. The oven interior shall be provided with a light which will be recessed and protected with a heat-resistant window. The light shall be activated by either a manual switch or an automatic door switch.

3.25 Accessibility. The unit shall be physically and visually accessible from the front for operation and for normal maintenance with tools, test equipment, and replacement parts. No special tools shall be required for access or parts replacement.

3.26 Physical size limitations and modularity. Modules of the disassembled size 1 range shall pass through a shipboard access measuring 26 x 66 inches with 8-inch radius corners. Modules of the disassembled size 2 range shall pass through a 30 inch diameter submarine access. (Shipboard requirement, see 7.2)

3.27 Label plates. The unit shall be provided with a data nameplate and an instruction plate. They shall be attached to the unit, readily visible during normal operating use, and shall not adversely affect the life and utility of the unit.

3.27.1 Data nameplate. The data nameplate shall contain the manufacturer’s name, model, serial number, date manufactured, and any other information needed to uniquely identify the unit.

3.27.2 Instruction plate. The instruction plate shall provide instructions for start-up, operation, and shut down.

3.28 High voltage labels. A label reading “Danger—High Voltage” shall be affixed to the outer case assembly, on or adjacent to each service access cover near one of the fasteners securing the cover. In addition, a warning label in accordance with UL 969 shall be placed near the high voltage components inside the equipment. This label shall include, but not be limited to the following texts:

- Danger—High Voltage.
- Power supply must be disconnected before servicing.
- Access covers must be in place before use.
- Service should be performed by authorized personnel only.

3.29 Standards compliance. The unit shall be in accordance with and certified to meet the applicable requirements of NFPA 70, NSF 4, and UL 197.

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4. REGULATORY REQUIREMENTS

The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation.

5. QUALITY ASSURANCE PROVISIONS

5.1 Product conformance. The product provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market, or the same product that has been delivered to the Government for shipboard use on a previous procurement. The Government reserves the right to require proof of such compliance.

6. PACKAGING

Preservation, packing, and marking shall be as specified in the contract or purchase order.

7. NOTES

7.1 Ordering data.

- Title, number, and date of this CID
- Style and size required
- French or regular hot plates
- When required, manuals shall be in accordance with ASTM F 760

7.2 Shipboard requirement. Whenever a "(Shipboard requirement)" is included in a paragraph under SALIENT CHARACTERISTICS, it is meant that the requirement is something that is not normally offered to the commercial market by the manufacturer.

7.3 Source of documents.

7.3.1 Military documents. Copies of documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.

MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally excited)
MIL-STD-1399 - Interface Standard for Shipboard Systems Section 300A Electric Power, Alternating

7.3.2 American Society for Testing and Materials (ASTM) Standards. ASTM Standards are available from the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohoken, PA 19428-2959.

ASTM A 167 - Stainless and Heat-Resisting Chromium-Nickel Steel Plate, Sheet, and Strip
ASTM F 760 - Food Service Equipment Manuals

7.3.3 National Sanitation Foundation International (NSF) Standards. NSF Standards are available from the National Sanitation Foundation International, 3475 Plymouth Road, P.O. Box 130140, Ann Arbor, MI 48113-0140.

NSF 4 - Commercial Cooking, Rethermalization and Powered Hot Food Holding and Transport Equipment

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7.3.4 National Fire Protection Association (NFPA) Standards. NFPA Standards are available from the National Fire Protection Association, 1 Batterymarch Park, Quincy, MA 02269-9101.

NFPA 70 - National Electrical Code

7.3.5 Underwriters Laboratories (UL) Standards. UL Standards are available from the Underwriters Laboratories Inc., 333 Pfingston Road, Northbrook, IL 60062.

UL 197 - Commercial Electric Cooking Appliances
UL 969 - Marking and Labeling Systems

7.3.6 The Institute of Electrical and Electronic Engineers (IEEE) Standards. IEEE Standards are available from The Institute of Electrical and Electronic Engineers, 445 Hose Lane, P.O. Box 1331, Piscataway, NJ 08855-1331.

IEEE 45 - IEEE Recommended Practice for Electric Installations on Shipboard

7.3.7 The National Electrical Manufacturers Association (NEMA) Standards. NEMA Standards are available from the National Electrical Manufacturers Association, 1300 North 17th Street, Suite 1847, Rosslyn, VA 22209.

NEMA 250 - Enclosures for Electrical Equipment (1000 Volts Maximum)

7.4 Suggested sources of supply. Manufacturers of products known to meet the requirements of this CID are listed below. However, competition is not limited to these companies.

Alloy Metal Products Inc.
14846 Persistence Dr.
Woodbridge, VA 22192

Lang Manufacturing Co.
6500 Merrill Creek Parkway
Everett, WA 98201

Vulcan-Hart Corp.
3600 North Point Blvd.
Baltimore, MD 21222-2726

8. CONCLUDING MATERIAL

Custodian:
Navy - SH

Preparing activity:
Navy - SH
(Project 7310-0896)

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